MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL SEMESTER 2025
CHAPTER 2: SEQUENCES (PART 1)

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Goals of this section:

(1) Sequences and convergence
(2) Monotone sequences and MCT
(3) Examples

Reference books:

- Robert G. Bartle, Donald R. Sherbert: Introduction to Real Analysis (Wiley, 2011, Fourth Edition)
- Kenneth A. Ross: Elementary Analysis: The Theory of Calculus (Springer, 2013, Second Edition)

1. Sequence of numbers and convergence

1.1. **Definition.**

Definition 1.1. A sequence in $\mathbb{R}$ is a function x from $\mathbb{N}$ to $\mathbb{R}$. We shall denote the element $x(n)$ by x_n for $n \in \mathbb{N}$. We also sometimes write a sequence of the form $(x_1, x_2, \ldots)$ or (x_n). For example, if we write $(2, 4, 6, 8, \ldots)$, then it means the sequence is that $x_1 = 2, x_2 = 4, x_3 = 6, x_4 = 8, \ldots$.

Example 1.2.
(a) Let $x_n = 3$ for all $n \in \mathbb{N}$. Then the sequence is the same for all $n \in \mathbb{N}$, and it takes the form $(3, 3, 3, \ldots)$.
(b) Let $r \in \mathbb{R}$. Let $x_n = r^n$. This is so called the geometric sequence.
(c) Let $x_n = (-1)^n$. The sequence take the form $(-1, 1, -1, 1, \ldots)$.
(d) The celebrated Fibonacci sequence (f_n) is defined as:

$$f_1 := 1, \quad f_2 := 1, \quad f_{n+1} := f_n + f_{n-1} \ (n \geq 2)$$

Exercise 1.3.
(1) Let (x_n) be a sequence in $\mathbb{R}$. Explain that there is a number r in $\mathbb{R}$ such that $x_n \neq r$ for all $n \in \mathbb{N}$
(2) Describe a seqeunce x_n in $\mathbb{Q}$ such that for each number r in $\mathbb{Q}$, there is a unique $m \in \mathbb{N}$ such that $x_m = r$.

1.2. **Convergence.**

Definition 1.4. (1) A sequence (x_n) in $\mathbb{R}$ is said to converge to x in $\mathbb{R}$, or $x \in \mathbb{R}$ is said to be the limit of a sequence (x_n) if for every $\epsilon > 0$, there exists $N(\epsilon)$ such that for any $n > N(\epsilon)$, $|x_n - x| < \epsilon$. (*We may sometimes simply write N rather than $N(\epsilon)$.*)

(2) If a sequence has a limit, we say that the sequence is convergent. If the sequence has no limit, then we say that the sequence is divergent.

(3) When a sequence in $\mathbb{R}$ has the limit x, we shall write

$$\lim x_n = x \quad \text{or} \quad \lim_{n\to\infty} x_n = x.$$

Exercise 1.5. Draw a real line and indicate which x_n lies in the interval $(x-\epsilon, x+\epsilon)$ in Definition 1.4.

1.3. **Uniqueness of convergence.**

Theorem 1.6. *(Uniqueness of a limit) Each convergent sequence has unique limit.*

Proof. Let (x_n) be a convergent sequence in S. Let x' and x'' be limits of (x_n). Let $\epsilon = \frac{|x'-x''|}{2}$. Suppose $x' \neq x''$ and so $\epsilon > 0$ to derive a contradiction.

Step 1: Large n so that x_n is close to x' and x''

By the definition of limits, there exist integers $N'(\epsilon)$ such that for $n > N'(\epsilon)$,

$$|x_n - x'| < \epsilon$$

Similarly, there exist integers $N''(\epsilon)$ such that for $n > N''(\epsilon)$,

$$|x_n - x''| < \epsilon$$

Step 2: Get contradiction that for large n, x_n cannot be so close to both x' and x''

Now, pick $n^* > \max\{N'(\epsilon), N''(\epsilon)\}$. Then,

$$|x' - x''| = |x' - x_{n^*} + x_{n^*} - x''| \leq |x' - x_{n^*}| + |x_{n^*} - x''| < \epsilon + \epsilon = 2\epsilon.$$

However, this means $|x' - x''| < |x' - x''|$, giving a contradiction. Thus, we must have that $x' = x''$. □

Example 1.7. (a) Guess $\lim_{n\to\infty}(\frac{1}{n})$ and then use the $\epsilon - N$-definition to show that.

Sol: Given $\epsilon > 0$, we find a number $N > \frac{1}{\epsilon}$. Then, for any $n > N$,

$$|\frac{1}{n} - 0| = \frac{1}{n} < \frac{1}{N} < \epsilon.$$

By the definition of a limit, $\lim(\frac{1}{n}) = 0$.

(b) Guess $\lim(\frac{2}{n^2+2})$ and then use $\epsilon - N$ definition to show that.
Sol: Given $\epsilon > 0$, we solve the equation $\frac{2}{N^2+2} < \epsilon$. Indeed,

$$\frac{2}{N^2+2} < \epsilon \Longrightarrow N^2 + 2 > \frac{2}{\epsilon}$$

- If $\epsilon \geq 1$, then we can choose any N, say $N = 1$. Then, for any $n > 1$, $\frac{2}{n^2+2} < \frac{2}{2} = 1$ and so $|\frac{2}{n^2+2} - 0| < 1 \leq \epsilon$.
- If $\epsilon < 1$, then we choose $N > \sqrt{\frac{2}{\epsilon} - 2}$, and so

$$N^2 > \frac{2}{\epsilon} - 2 \Rightarrow \epsilon > \frac{2}{N^2+2}.$$

Then, for any $n > N$,

$$\frac{2}{n^2+2} < \frac{2}{N^2+2} < \epsilon \Rightarrow |\frac{2}{n^2+2} - 0| < \epsilon.$$

Exercise 1.8. (Example of non-convergent sequence) Let $x_n = (-1)^n$. Show that (x_n) is divergent by the $\epsilon - N$ definition of a limit.

Exercise 1.9. (Tail of sequences) Let (x_n) be a sequence in $\mathbb{R}$. Let $m \in \mathbb{N}$. Define a new sequence $y_n = x_{m+n}$. Use the $\epsilon - N$-argument (x_n) is convergent if and only if (y_n) is also convergent. Moreover, if one of the limits exist, then $\lim y_n = \lim x_n$.

In the statements below (e.g. Lemma 2.4 and Theorem 3.6), we deal with 'for all $n \in \mathbb{N}$', but one can use the 'tail sequence' to replace the condition 'there exists $m \in \mathbb{N}$ such that for all $n > m$.

2. Boundedness of a convergent sequence

2.1. Convergnet $\Rightarrow$ Bounded.

Definition 2.1. A sequence (x_n) in $\mathbb{R}$ is said to be bounded if the set $\{x_n : n \in \mathbb{N}\}$ is bounded i.e. there exists numbers L, M in $\mathbb{R}$ such that $L < x_n < M$ for all $n \in \mathbb{N}$.

Theorem 2.2. *Any convergent sequence in $\mathbb{R}$ is bounded.*

Proof. Let (x_n) be a convergent sequence, and let $x = \lim x_n$. Pick $\epsilon = 1$. Then there exists an integer $N \in \mathbb{N}$ such that for any $n > N$, $|x_n - x| < 1$ and so $x_n < x + 1$. Let $M = \max\{x + 1, x_1, \ldots, x_N\}$. Then $x_n \leq M$ for any $n \in \mathbb{N}$, and hence $\{x_n : n \in \mathbb{N}\}$ is bounded above.

The argument for bounded below is similar. □

Example 2.3. Let $x_n = \sqrt{n + \frac{1}{n}}$. The sequence (x_n) is not bounded and so is divergent.

2.2. **Bounding a convergent sequence.**

Lemma 2.4. *Let (x_n) be a convergent sequence in $\mathbb{R}$. If there exist $L, M \in \mathbb{R}$ such that $L \leq x_n \leq M$ for all $n \in \mathbb{N}$, then $L \leq \lim x_n \leq M$.*

Proof. Let $x = \lim x_n$. Suppose $x > M$. Let $\epsilon > 0$ such that $x - \epsilon > M$ (why such ϵ exists?). By definition of limit, there exists $N \in \mathbb{N}$ such that for any $n > N$,

$$|x_n - x| < \epsilon$$

and so $x - \epsilon < x_n < x + \epsilon < 0$.

The last inequality implies there exists $n^* \in \mathbb{N}$ such that $x_{n^*} > x - \epsilon > M$. This gives a contradiction. Hence $x \leq M$. Proving $L \leq x$ is similar. □

The following reformulation is useful later:

Corollary 2.5. *Let (x_n) be a convergent sequence in $[a, b]$. Then* $\lim\ x_n \in [a, b]$.

Exercise 2.6. (a) Does the converse of Theorem 2.2 hold?

(b) Could we replace the non-strict inequalities in Lemma 2.4 by strict inequalities?

3. Some basic methods of showing convergence

3.1. **Construct convergent sequences by basic operations.** We shall explain how to use some simple operations such as additions and multiplications to obtain convergence from known ones.

Theorem 3.1. *Let (x_n) and (y_n) be two convergent sequences in $\mathbb{R}$. Let $x = \lim x_n$ and let $y = \lim y_n$.*

(a) *For $c \in \mathbb{R}$, (cx_n) converges to cx.*
(b) *$(x_n + y_n)$ converges to $x + y$.*
(c) *$(x_n y_n)$ converges to xy.*
(d) *$(-x_n)$ converges to $-x$.*
(e) *$x_n - y_n$ converges to $x - y$.*
(f) *If $x_n \neq 0$ for all $n \in \mathbb{N}$ and $\lim x_n \neq 0$, then $\lim \frac{1}{x_n} = \frac{1}{x}$.*

Proof. For any $\epsilon > 0$, there exists an integer $N(\epsilon) > 0$ such that for any $n > N(\epsilon)$,

$$|x_n - x| < \epsilon, \text{ and } |y_n - y| < \epsilon.$$

Prove (a). Let $\epsilon > 0$. If $c = 0$, a constant sequence is always convergent and (cx_n) converges to 0. Suppose $c \neq 0$. Then, for $n > N(\frac{\epsilon}{|c|})$,

$$|cx_n - cx| = |c||x_n - x| < |c|\frac{\epsilon}{|c|} = \epsilon.$$

Prove (b). Let $\epsilon > 0$. Then, for $n > N(\frac{\epsilon}{2})$,

$$|(x_n + y_n) - (x + y)| = |(x_n - x) + (y_n - y)| \leq |x_n - x| + |y_n - y| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Prove (c). *Step 1: Bound sequences.* Let $\epsilon > 0$. Then, by Theorem 2.2, (x_n) and (y_n) are bounded, and so there exists a number M in $\mathbb{R}$ such that $|x_n| < M$ and $|y_n| < M$.
Step 2: Bound limits By Lemma 2.4, $|y| \leq M$ (we also have $|x| \leq M$, but we do not need this).
Step 3: Form inequalities Then, for $n > N(\frac{\epsilon}{2M})$,

$$|x_n y_n - xy| = |x_n y_n - x_n y + x_n y - xy| \leq |x_n y_n - x_n y| + |x_n y - xy| < M|y_n - y| + M|x_n - x| = \epsilon.$$

Prove (d). Let $\epsilon > 0$. Then, for any $n > N(\epsilon)$,

$$|(-x_n) - (-x)| = |-x_n + x| = |x_n - x| < \epsilon.$$

Prove (e). This follows from (b) and (d).

Prove (f).(Sketch) *Step 1:* First show that there exists a real number $M > 0$ such that $|x_n| > M$. (Hint: Similar argument to the proof of Theorem 2.2)

Step 2: Show that $|x| \geq M$ by Theorem 2.4.

Step 3: Let $\epsilon > 0$. Then, for any $n > N(M^2\epsilon)$,

$$\left|\frac{1}{x_n} - \frac{1}{x}\right| = \frac{|x - x_n|}{|x_n||x|} \leq \frac{1}{M^2}|x - x_n| < \epsilon.$$

□

Example 3.2. Let $x_n = 2\left(1 - \frac{2}{n}\right)$ for all $n \in \mathbb{N}$. Find $\lim x_n$.
Sol: It follows from Theorem 3.1 that

$$\lim x_n = 1 - 2\lim \frac{1}{n} = 1 - 2(0) = 1.$$

Exercise 3.3. Let k be a positive integer. Let (x_n) be a sequence converging to x. Show that $\lim x_n^k = x^k$.

Exercise 3.4. Let $x_n = \frac{n^2+2n+3}{3n^2+4n+5}$. Find $\lim\ x_n$.

Exercise 3.5. Let $(x_n), (y_n)$ and (z_n) be convergent sequences. Suppose

$$x_n \leq y_n \leq z_n$$

for all $n \in \mathbb{N}$. Then

$$\lim\ x_n \leq \lim\ y_n \leq \lim\ z_n.$$

3.2. **Squeeze theorem.**

Theorem 3.6. *(Squeeze theorem) Let $(x_n), (y_n), (z_n)$ be three sequences in $\mathbb{R}$. Suppose*

$$x_n \leq y_n \leq z_n$$

for all $n \in \mathbb{N}$. Suppose x_n and z_n are both convergent and $\lim x_n = \lim z_n$. Then (y_n) is also convergent and $\lim y_n = \lim x_n$.

Proof. Let $w = \lim\ x_n$. Given $\epsilon > 0$, there exists an integer N such that for all $n > N$,

$$|x_n - w| < \epsilon, \quad |z_n - w| < \epsilon$$

and so $-\epsilon < x_n - w < \epsilon, \quad -\epsilon < z_n - w < \epsilon$.

On the other hand,

$$x_n - w \leq y_n - w \leq z_n - w$$

and so $-\epsilon < y_n - w < \epsilon$. Thus, $|y_n - w| < \epsilon$. □

Example 3.7. $\lim(\sqrt{n+1} - \sqrt{n}) = 0$. Note that

$$\lim\ (\sqrt{n+1} - \sqrt{n}) = \lim\ \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

Note that, for all $n \in \mathbb{N}$,

$$0 < \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{2\sqrt{n}}$$

Since $\lim \frac{1}{2\sqrt{n}} = 0$, the sequeeze theorem says that

$$\lim\ \frac{1}{\sqrt{n+1} + \sqrt{n}} = 0$$

and so $\lim\ (\sqrt{n+1} - \sqrt{n}) = 0$.

Example 3.8. See the book of Bartle-Sherbert (f) Page 67.

Exercise 3.9. Let (x_n) be a sequence in $\mathbb{R}$. Let $x \in \mathbb{R}$. Let (a_n) be a sequence of positive numbers such that $\lim a_n = 0$. Suppose, for all $n \in \mathbb{N}$,

$$|x_n - x| \leq a_n.$$

Show that $\lim\ x_n = x$.

4. Monotone Convergence Theorem

4.1. Monotone sequences.

Definition 4.1. A sequence in $\mathbb{R}$ is said to be increasing if it satisfies the inequality:

$$x_1 \leq x_2 \leq x_3 \leq \dots$$

A sequence in $\mathbb{R}$ is said to be decreasing if it satisfies the inequality:

$$x_1 \geq x_2 \geq x_3 \geq \dots$$

A sequence is said to be monotone if it is either increasing or decreasing.

Example 4.2.
(1) The sequence (2^n) is increasing.
(2) The sequence $(n+2)$ is increasing.
(3) The sequence $(\frac{1}{2^n})$ is decreasing.
(4) The sequence $(\frac{1}{n^2+2})$ is decreasing

4.2. A bounded criteria.

We have the following criteria for the convergence of a monotone sequence:

Theorem 4.3. *(Monotone Convergence Theorem) A monotone sequence is convergent if and only if it is bounded.*

Proof. We shall consider the sequence to be increasing. The situation for a decreasing sequence is similar.

If direction:

Suppose the sequence (x_n) is bounded. Let $s = \sup(x_n)$, which exists by the completeness property. Then for any given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $s - \epsilon < x_N$. Since the sequence is increasing, for any $n > N$,

$$x_n \geq x_N > s - \epsilon.$$

Thus, $s - x_n < \epsilon$ and so $|s - x_n| < \epsilon$. By definition, s is the limit of (x_n).

Only if direction:

This follows from Theorem 2.2. □

The proof of Theorem 4.3 gives that:

Corollary 4.4.
(1) *If (x_n) is a bounded increasing sequence, then*

$$\lim(x_n) = \sup\{x_n : n \in \mathbb{N}\}.$$

(2) *If (x_n) is a bounded decreasing sequence, then*

$$\lim(x_n) = \inf\{x_n : n \in \mathbb{N}\}.$$

Remark: The above corollary does not hold if one removes the condition of monotone. For example, take $x_1 = 3$, for $n \geq 2$, $x_n = 2 - \frac{1}{n}$. Then the sequence is convergent and $\lim\ x_n = 2$, but $\sup\{x_n : n \in \mathbb{N}\} = 3$.

4.3. **Examples and applications.**

Example 4.5. Let $1 < x_1 < 2$. Let $x_{n+1} = 1 + \sqrt{x_n - 1}$. Show that $\lim x_n$ exists in $\mathbb{R}$ and find it.

Proof: We shall show that $x_n < x_{n+1}$ for all $n \in \mathbb{N}$ and $1 < x_n < 2$. We shall proceed by the mathematical induction.

For x_1, it follows from the given condition. Suppose the statement is true for certain $m \in \mathbb{N}$. Since $x_m - 1 < 1$, $0 < \sqrt{x_m - 1} < 1$ and so $0 < x_m - 1 < \sqrt{x_m - 1}$. Thus, $x_m - 1 < x_{m+1} - 1$ and so $x_m < x_{m+1}$. This then also implies that $1 < x_{m+1}$ since $1 < x_m$. We also have $x_{m+1} < 2$ since $x_m - 1 < 1 \Rightarrow \sqrt{x_m - 1} < 1 \Rightarrow 1 + \sqrt{x_m - 1} < 2$.

Since (x_n) is also bounded, the monotone convergence theorem also implies that $\lim x_n$ exists. Let $L = \lim x_n$. Then

$$\lim(x_{n+1} - 1)^2 = \lim(x_n - 1)$$

and so

$$(\lim\ x_{n+1} - 1)^2 = \lim\ x_n - 1.$$

Hence, $(L-1)^2 = (L-1)$ and so $(L-1)(L-2) = 0$. Then, $L = 1$ or 2. Since $1 < x_1 \leq L$, we can only have $L = 2$.

Exercise 4.6. Define $x_1 = \sqrt{3}$. Define $x_n^2 = 3x_{n-1}$ for $n \geq 2$. Show that $\lim x_n$ exists and find it.

5. Some useful limits

5.1. **Geometric Sequence.**

Proposition 5.1. (1) *If* $0 < r < 1$, *then* $\lim_{n\to\infty} r^n = 0$.

(2) *If* $r > 1$, *then* (r^n) *is divergent.*

Proof. (1) Since $r^{n+1} = r(r^n) < r^n$, the sequence $\{r^n\}$ is decreasing and so monotone. Since $\{r^n\}$ is bounded, the MCT implies that the sequence is convergent. Let $x_n = r^n$. Then $x_{n+1} = rx_n$ and so $\lim(x_{n+1}) = \lim(rx_n)$. Then, $L = rL$ and so $L = 0$.

(2) Let $x_n = r^n$. If $\lim_{n\to\infty} x_n$ exists, let $L = \lim_{n\to\infty} x_n$. Since $x_{n+1} = rx_n$, $\lim_{n\to\infty} x_{n+1} = r\lim_{n\to\infty} x_n$ and so $L = rL$. Thus, $L = 0$. But $x_n > 1$ for all n and $L \geq 1$. This gives a contradiction.

□

5.2. $n^{1/n}$.

Theorem 5.2.

$$\lim n^{1/n} = 1.$$

Proof. Let $x_n = n^{1/n} - 1$. For $n > 1$, $x_n > 0$. We now consider

$$n = (x_n + 1)^n$$

and so

$$n = 1 + nx_n + \frac{n(n-1)}{2}x_n^2 + \ldots > 1 + \frac{n(n-1)}{2}x_n^2.$$

Hence, $(n-1) > \frac{n(n-1)}{2}x_n^2$ and so

$$\frac{2}{n} > x_n^2.$$

This gives $\sqrt{\frac{2}{n}} > x_n > 0$. Since $\lim \sqrt{\frac{2}{n}} = 0$, by Squeeze Theorem,

$$\lim x_n = 0.$$

Hence, $\lim n^{1/n} = \lim\ (x_n + 1) = 1$. □

Exercise 5.3. Find the limits of the following sequences:

(1) $\lim\ n^{1/n^3}$
(2) $\lim\ (n!)^{1/n}$
(3) For $a > 0$, $\lim\ a^{1/n}$.

5.3. **Euler number.**

Theorem 5.4. *The sequence $\left(1 + \frac{1}{n}\right)^n$ converges. The limit will be denoted by the number e.*

Proof. Define $x_n = \left(1 + \frac{1}{n}\right)^n$. Then,

$$\begin{aligned} x_n &= \left(1 + \frac{1}{n}\right)^n \\ &= 1 + \frac{n}{1} \cdot \frac{1}{n} + \frac{n(n-1)}{2} \cdot \frac{1}{n^2} + \frac{n(n-1)(n-2)}{3!} \cdot \frac{1}{n^3} + \ldots + \frac{n(n-1)\ldots 1}{n!} \cdot \frac{1}{n^n} \\ &= 1 + 1 + \frac{1}{2}\left(1 - \frac{1}{n}\right) + \frac{1}{3!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) + \ldots + \frac{1}{n!} \cdot \left(1 - \frac{1}{n}\right)\ldots\left(1 - \frac{n-1}{n}\right) \end{aligned}$$

Increasing: Note that x_{n+1} has similar expressions and the middle terms look like:

$$\frac{1}{k!}\left(1 - \frac{1}{n+1}\right)\ldots\left(1 - \frac{k-1}{n+1}\right)$$

which is greater than the corresponding term

$$\frac{1}{k!}\left(1 - \frac{1}{n}\right)\ldots\left(1 - \frac{k-1}{n}\right)$$

in x_n. Moreover, x_{n+1} has one more term than x_n, and so $x_{n+1} > x_n$. This shows that (x_n) is increasing.

Bounded above: Note that

$$\frac{1}{k!}\left(1-\frac{1}{n}\right)\cdots\left(1-\frac{k-1}{n}\right) \leq \frac{1}{k!} \leq \frac{1}{2^k}$$

Thus,

$$x_n \leq 1+1+\frac{1}{2}+\frac{1}{2^2}+\ldots < 3.$$

Conclusion:

By the monotone convergence theorem, the sequence (x_n) converges. □

Exercise 5.5. Find the limit of the following sequences:

(1) $\left(1+\frac{1}{n+1}\right)^n$;

(2) $\left(1+\frac{1}{n}\right)^{2n}$.

5.4. $n\sin\left(\frac{1}{n}\right)$.

Exercise 5.6. Find, by using trigonometry, that

$$\lim_{n\to\infty} n\sin\left(\frac{1}{n}\right)$$